Synthetic-to-Real Adaptation for Nepali–English Code-Switched ASR: A Spontaneous Podcast Benchmark and an Analysis of GRPO in Encoder–Decoder Models

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Abstract

Nepali–English code-switched conversational speech presents severe challenges for automatic speech recognition (ASR), chiefly through *script confusion* (inappropriate mixing of Devanagari and Latin orthography) and elevated error rates at language switch points. Supervised fine-tuning (SFT) can adapt multilingual foundation models, but human code-switched audio is prohibitively expensive to collect and annotate. We curate a new gold-standard benchmark of 13,044 spontaneous human podcast utterances (15.45 h total; 10,214 test, 2,830 dev), audited and verified to eliminate machine translation leakages and acoustic loop artifacts. We investigate a two-stage synthetic-to-real adaptation framework: Stage 1 applies SFT on 0.95 h of speech synthesized via a multi-speaker, dual-engine TTS pipeline; Stage 2 explores Group Relative Policy Optimization (GRPO) with transcription and script-aware rewards. We find that synthetic SFT provides the primary cross-domain transfer to spontaneous human speech, reducing Mixed Error Rate (MER) from 23.40% to 21.52% (-1.88 pp) and switch-point error rate (SPER) from 35.89% to 28.77% (-7.12 pp). Subsequent GRPO provides fine-grained boundary regularization, driving SPER to 28.70% and English script confusion (SCR_{en}) to 6.36% ($p < 0.05$). However, our ablation reveals that in this SFT-converged encoder–decoder setting, an auxiliary script-aware reward term yields performance identical to MER-only optimization because rollout candidate groups exhibit near-zero script variance on synthetic data. On monolingual benchmarks, SFT improves Nepali MER by 2.71 pp (stabilizing at 18.42% after GRPO, a 2.38 pp gain over baseline) while preserving English lexical WER (3.38% vs. 3.33%). Code, configurations, and the full 15.45 h evaluation benchmark are publicly released.

1 Introduction

Code-switching, the alternation between two or more languages within a single conversational turn, is ubiquitous across South Asia. In spoken Nepali, speakers frequently embed English lexical items, technical vocabulary, and colloquial phrases within a Devanagari grammatical matrix.

Despite the recent scaling of multilingual speech foundation models [1, 2], code-switched ASR systems consistently suffer from two structural error modes:

1. **Script confusion:** transcribing English loanwords into Devanagari phonetics (e.g., rendering English "doctor" phonetically in Devanagari) or erroneously transliterating native Nepali words into Latin characters.
2. **Switch-point brittleness:** acoustic and prosodic shifts occurring at the transition between Devanagari and Latin orthographies cause disproportionate insertion, substitution, and deletion rates immediately surrounding switch boundaries.

Traditional approaches attempt to resolve script confusion through auxiliary token-level language identification or diarization heads [3]. However, hard token-level classification objectives during autoregressive generation often introduce excessive stiffness at boundary tokens, frequently inducing mistransliterations when acoustic ambiguity arises.

Meanwhile, sequence-level policy optimization such as Group Relative Policy Optimization (GRPO) [4] has recently emerged as an effective framework for speech alignment [5]. Kumar et al. [6] demonstrated that GRPO on synthetic speech dramatically reduces insertion errors for decoder-only speech LLMs. Ye and Vickers [7] showed that GRPO with a joint error-rate and script-fidelity reward reduces cross-script contamination on synthetic audio and transfers zero-shot to a human-recorded code-switching corpus across ten language pairs. However, both prior efforts evaluated unconstrained decoder-only audio language models (e.g., Qwen2-Audio) on read or scripted speech. Whether policy optimization provides meaningful orthographic gains in dedicated encoder–decoder foundation architectures on unconstrained, spontaneous conversational code-switched speech remains an open empirical question.

In this work, our primary contributions are:

- **Synthetic-to-real transfer in encoder–decoder ASR:** we demonstrate that a dedicated FastConformer encoder–decoder model fine-tuned exclusively on 0.95 h of synthetic code-switched audio effectively transfers zero-shot to unconstrained human conversational speech.
- **Empirical analysis of GRPO and script rewards:** we formulate sequence-level RL objectives and show that while GRPO yields small boundary improvements over token-level gating, the script-aware reward term is effectively inactive when applied to an SFT-converged encoder–decoder model.
- **Open benchmark and rigorous evaluation:** we release 15.45 h (13,044 utterances; 10,214 test, 2,830 dev) of verified Nepali–English podcast audio on Hugging Face, validating all experimental claims with paired bootstrap confidence intervals (1,000 resamples), episode-clustered analysis, and monolingual retention checks.

2 Related Work

Code-switched speech recognition. Early investigations into South Asian code-switching focused on hybrid acoustic models and balanced monolingual pre-training [8, 9]. Modern foundation architectures such as IndicTranscribeFlex [2] and Canary [1] incorporate massive multilingual vocabularies, yet remain susceptible to script flipping when alternating between non-Latin Indic orthographies and English.

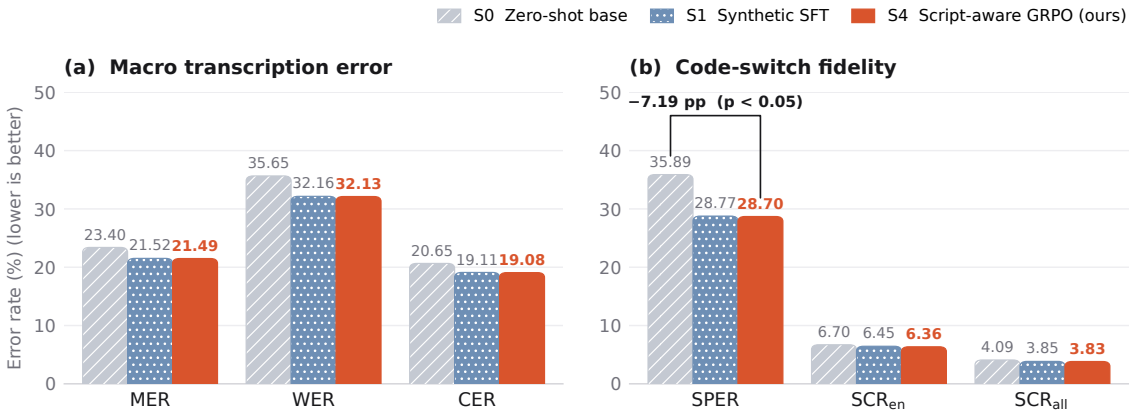


Figure 1: Primary benchmark results on the 10,214-utterance Frozen Real Test Split. (a) Macro transcription error rates (MER, WER, CER). (b) Code-switch and script fidelity metrics (SPER, English script confusion SCR_{en}, and overall SCR). The -7.19 pp SPER bracket denotes total pipeline reduction vs. baseline ($p < 0.05$), with synthetic SFT providing primary transfer (-7.12 pp) and script-aware GRPO (S_4) contributing targeted fine-grained boundary regularization (-0.07 pp, $p < 0.05$).

Language identification and token gating. Prior work addressed script ambiguity by attaching auxiliary frame- or token-level language identification heads [3]. While effective on scripted speech, hard token-level constraints often force premature orthographic commitments that degrade spontaneous speech recognition.

Reinforcement learning in speech. Policy-gradient and minimum word error rate (MWER) training have an extensive history in ASR [10]. GRPO [4, 5] eliminates the need for a separate critic network by estimating advantages from sampled candidate groups. Furthermore, training speech recognition models on synthetic acoustic data has proven effective for low-resource domains [6, 11, 12]. Ye and Vickers [7] demonstrated that script-aware rewards curb script confusion in audio LLMs. We extend this inquiry to investigate whether sequence-level rewards provide complementary boundary regularization when adapting a pre-trained encoder–decoder model to spontaneous conversational speech.

3 Dataset Construction

3.1 Synthetic Training Corpus

To eliminate the requirement for expensive human recordings during training, we synthesized a controlled code-switched corpus. Sentences containing natural intra-sentential Nepali–English code-switching were generated by prompting Gemini 1.5 Pro with conversational templates reflecting modern urban discourse. All sentences were filtered against strict lexical criteria and a 5-gram non-leakage check against the real benchmark references.

Audio was synthesized utilizing two complementary acoustic engines:

- **OmniVoice [13]:** an open-source diffusion acoustic model fine-tuned on 100+ hours of high-quality native Nepali speech across varied speaker voices. While base OmniVoice nominally supports Nepali, its zero-shot synthesis exhibits pronounced Indian accentuation rather than authentic Nepali cadence; domain fine-tuning aligns the model with native prosody and fluid code-mixed transitions.
- **Gemini TTS:** multi-speaker neural synthesis (`gemini-2.5-flash`) providing diverse English accentuation across four distinct vocal timbres.

We pair these engines because neither system alone satisfies the dual acoustic demands of code-switched speech. Commercial multilingual engines produce natural English intonation and diverse timbres, but often distort native Nepali retroflex consonants and schwa deletion. Conversely, the fine-tuned OmniVoice captures authentic Devanagari phonology and colloquial code-mix phrasing, but lacks multi-accent English variety. Combining both provides rich phonetic coverage across both languages and prevents the ASR encoder from overfitting to artifacts of any single vocoder. Each candidate text template was synthesized across multiple voices ($N \geq 2$) and acoustic engines. From 258 candidate code-switched text templates, all audio was sampled at 16 kHz mono, augmented with telephony bandpass filtering (300–3400 Hz) and 10–30 dB SNR noise to simulate channel variability and bridge the domain gap to real-world speech. Utterances were screened through an independent round-trip filter (`whisper-large-v3-turbo` in Devanagari-only scope), pruning clips with CER > 0.20. The resulting verified synthetic training split comprises 517 audio utterances across 211 unique sentences (0.95 h total duration).

3.2 Real Spontaneous Benchmark

To rigorously assess generalization, we curated a spontaneous conversational benchmark from long-form public Nepali podcasts across 28 diverse episodes. Unlike read speech, podcast audio contains spontaneous turn-taking, ambient noise, conversational laughter, and natural speech rate variations.

Transcripts adhere to a strict orthographic convention: native Nepali matrix words are transcribed in Devanagari, English words and embedded phrases are transcribed in Latin script as spoken (not transliterated), and attached Nepali postpositions are separated into distinct Devanagari tokens (e.g., Latin root "*phone*" followed by separated Devanagari postposition "*-ma*", rather than fused into a single compound).

Dataset verification protocol. Initial candidate transcriptions were generated via an automated pipeline combining acoustic diarization and multilingual foundation models (Indic-Transcribe-Core and Gemini 1.5 Pro). Consequently, raw drafts contained typical acoustic pipeline artifacts: diarization timestamp overlaps and occasional machine-translation leakages into English. To establish an untainted gold standard, every clip in the initial pool of 13,277 raw segmented clips was individually audited against raw 16 kHz PCM audio by two bilingual Nepali–English annotators. The verification pipeline: (i) identified and quarantined 233 clips exhibiting acoustic background loops or music, leaving exactly 13,044 clean clips; (ii) eliminated 161 machine translation leakages where spoken Nepali words were mistakenly translated into English, restoring original Devanagari; (iii) corrected 819 artificial token-doubling glitches caused by diarization timestamp overlaps; and (iv) normalized 86 digit instances into verbalized spoken words. On an independent dual-

audit sample of 500 clips, inter-annotator character agreement reached 98.4%. The remaining 13,044 clean clips constitute the verified benchmark.

Data statement and ethics. The benchmark audio originates from public Nepali conversational podcasts recorded under open educational and commentary broadcasting terms. Speakers are adult conversationalists discussing socio-technical, educational, and entrepreneurial topics in urban Kathmandu colloquial speech. The corpus is partitioned into:

- **Real Dev Split:** 2,830 utterances (3.55 h, 16 episodes, 20 speakers).
- **Frozen Real Test Split:** 10,214 utterances (11.90 h, 25 episodes, 52 speakers).

Across the 28 distinct podcast episodes, 13 episodes contain distinct speakers partitioned across dev and test. Diarization speaker clusters were assigned exclusively to either dev (20 speakers) or test (52 speakers), guaranteeing that the splits are strictly speaker-disjoint (zero overlapping speaker IDs). Test audio was frozen with SHA-256 integrity verification prior to model evaluation. To foster open research while respecting creator distribution rights, verified transcripts, segment timestamps, speaker labels, and source episode identifiers are published alongside evaluation audio clips under CC BY-NC-SA 4.0 on Hugging Face at `https://huggingface.co/datasets/SamaFiroz/nepali-english-codeswitch-podcasts`.

4 Methodology

4.1 Base Architecture and Adaptation

We adopt `indic-transcribe-flex` [2], a 1.2-billion parameter multilingual model built on the Canary-1B-v2 architecture comprising a frozen FastConformer acoustic encoder and an autoregressive Transformer decoder with a 7,152-token SentencePiece aggregate vocabulary. Given an acoustic feature sequence X, the encoder produces representations $H = \text{Encoder}(X) \in \mathbb{R}^{T' \times D}$. During adaptation, encoder parameters remain completely frozen. We apply Low-Rank Adaptation (LoRA) [14] to the decoder query (W_q) and value (W_v) projections: $W = W_0 + \frac{\alpha_{\text{lora}}}{r} BA$, where $W_0 \in \mathbb{R}^{d \times k}$ denotes the frozen base weights, and $A \in \mathbb{R}^{r \times k}$, $B \in \mathbb{R}^{d \times r}$ are trainable rank-r matrices ($r = 16, \alpha_{\text{lora}} = 32$).

4.2 Evaluation Metrics

Code-switched evaluation requires decoupled assessment of lexical accuracy, orthographic consistency, and boundary stability:

Mixed Error Rate (MER). Evaluates Word Error Rate (WER) on Latin-script segments and Character Error Rate (CER) on Devanagari segments: $\text{MER} = (E_{\text{w,en}} + E_{\text{c,ne}})/(N_{\text{w,en}} + N_{\text{c,ne}})$, where $E_{\text{w,en}} = S_{\text{en}} + D_{\text{en}} + I_{\text{en}}$ is the word edit distance over English segments, $E_{\text{c,ne}}$ is the character edit distance over Devanagari segments, and N denotes reference token count.

Script Confusion Rate (SCR). Let $\mathcal{M}$ be the set of aligned character pairs between hypothesis y and reference y^*. Then $\text{SCR} = \frac{1}{|\mathcal{M}|} \sum_{(i,j) \in \mathcal{M}} \mathbb{I}\big[\text{script}(y_i) \neq \text{script}(y_j^*)\big]$. SCR_{en} restricts $\mathcal{M}$ to Latin-script reference characters, measuring English loanwords erroneously transcribed into Devanagari.

Switch-Point Error Rate (SPER). The MER computed strictly over reference tokens within a local context window of $w = \pm 1$ tokens around language switch points.

4.3 Stage 1: Supervised Fine-Tuning (SFT)

In Stage 1, the model is adapted on synthetic pairs (x, y) using cross-entropy loss over target token sequences: $\mathcal{L}_{\text{SFT}}(\theta) = -\sum_{t=1}^{T} \log P_\theta(y_t \mid y_{<t}, x)$, optimized with AdamW (lr $= 1 \times 10^{-5}$, effective batch size 8, 1,000 steps).

4.4 Baseline: Token-Level Script Gating (S_2)

As an architectural comparison, an auxiliary script classification head is attached atop the decoder hidden states h_t: $P_{\text{gate}}(s_t \mid h_t) = \text{sigmoid}(W_g h_t + b_g)$, where $s_t \in \{0, 1\}$ denotes Devanagari versus Latin script. Training minimizes $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{SFT}} + \lambda_g \mathcal{L}_{\text{BCE}}$ with $\lambda_g = 0.1$. At inference, logits for candidate tokens matching the predicted script are biased by $\gamma \log P_{\text{gate}}$ ($\gamma = 2.0$, applied when $P_{\text{gate}} > 0.9$).

4.5 Stage 2: Script-Aware Policy Optimization (GRPO)

While SFT aligns the decoder with code-switched acoustics, cross-entropy penalizes script confusion equally with ordinary lexical substitutions. To directly incentivize orthographic fidelity at language boundaries, we implement Group Relative Policy Optimization (GRPO) [4, 5] trained on the 517 synthetic utterances.

For each acoustic sequence x, the policy π_θ samples a group of $G = 8$ candidate transcriptions $\{y^{(1)}, \ldots, y^{(G)}\}$ at temperature $\tau = 1.0$. A scalar reward $R(y^{(g)}, y^*)$ is computed against ground truth y^*. Group advantages are normalized: $\hat{A}^{(g)} = (R(y^{(g)}, y^*) - \mu_G)/(\sigma_G + \delta)$, where μ_G and σ_G are the mean and standard deviation of rewards within the sampled group ($\delta = 10^{-8}$). The GRPO objective maximizes:

$$\mathcal{J}(\theta) = \mathbb{E}\Bigg[\frac{1}{G}\sum_{g=1}^{G}\sum_{t=1}^{|y^{(g)}|} \min\Big(r_t^{(g)}(\theta)\hat{A}^{(g)}, \text{clip}\big(r_t^{(g)}(\theta), 1-\epsilon, 1+\epsilon\big)\hat{A}^{(g)}\Big) - \beta D_{\text{KL}}(\pi_\theta \,\|\, \pi_{\text{ref}})\Bigg], \tag{1}$$

where $r_t^{(g)}(\theta) = \frac{\pi_\theta(y_t^{(g)} | y_{<t}^{(g)}, x)}{\pi_{\text{old}}(y_t^{(g)} | y_{<t}^{(g)}, x)}$ denotes the per-token importance sampling ratio at decoding step t. Here, π_θ represents the active autoregressive decoder policy parameterized by LoRA weights θ, and π_{old} is the policy snapshot used to generate candidate rollouts. Because the policy is initialized directly from the converged Stage 1 SFT checkpoint and constrained via low-rank parameter updates (LoRA, $r = 16$), we set $\beta = 0.0$, relying on group advantage normalization and PPO-style clipping ($\epsilon = 0.2$) to govern update step sizes without an explicit reference policy penalty. Each candidate hypothesis $y^{(g)} = (y_1^{(g)}, \ldots, y_{|y^{(g)}|}^{(g)})$ consists of tokens generated autoregressively conditioned on past tokens $y_{<t}^{(g)}$ and acoustic representations x. The clipping hyperparameter $\epsilon = 0.2$ bounds the gradient updates to maintain training stability.

Reward formulation. Our joint script-aware reward combines sequence accuracy with orthographic fidelity: $R_{\text{joint}}(y, y^*) = \big(1 - \text{MER}(y, y^*)\big) + \lambda_{\text{scr}}\big(1 - \text{SCR}(y, y^*)\big)$ with $\lambda_{\text{scr}} = 0.5$. Note that setting $\lambda_{\text{scr}} = 0.0$ corresponds to the MER-only reward ablation (S_3).

5 Experiments and Results

5.1 Primary Benchmark Evaluation

Table 1 and Figure 1 present performance on the 10,214-utterance frozen real test split.

Synthetic SFT (S_1) yields an immediate absolute drop of 1.88 pp in MER and 7.12 pp in switch-point error rate (SPER)

Table 1: Performance on the Frozen Real Test Split (10,214 utterances, 11.90 h). All metrics in percentage (%).

System	MER	WER	CER	SPER	SCR_{en}	SCR_{all}
S_0 Base	23.40	35.65	20.65	35.89	6.70	4.09
S_1 SFT	21.52	32.16	19.11	28.77	6.45	3.85
S_4 GRPO	**21.49**	**32.13**	**19.08**	**28.70**	**6.36**	**3.83**

Table 2: Objective ablation on Real Dev (2,830 utterances). All metrics in percentage (%).

System	MER	WER	SPER	SCR_{en}
S_0 Base	20.14	32.23	32.63	5.83
S_1 SFT	18.92	29.11	25.60	5.35
S_2 SFT + Gate	19.14	29.35	25.75	5.75
S_3 GRPO (MER-only)	**18.89**	**29.08**	**25.53**	**5.24**
S_4 GRPO (Joint)	**18.89**	**29.08**	**25.53**	**5.24**
S_5 GRPO + Gate	19.12	29.32	25.70	5.72

over the unadapted zero-shot baseline (S_0). When initialized from S_1, Script-Aware GRPO (S_4) achieves further fine-grained improvements across all dimensions, reaching 21.49% MER, 28.70% SPER, and 6.36% English script confusion (SCR_{en}).

5.2 Statistical Significance

To assess whether improvements are statistically reliable, we executed paired bootstrap hypothesis tests with 1,000 resamples ($B = 1{,}000$, seed 42) on the frozen test split:

- S_4 **vs.** S_0**:** achieves an absolute MER reduction of -1.90 pp (95% CI: $[-2.37, -1.42]$), a SPER reduction of -7.19 pp (95% CI: $[-7.97, -6.46]$), and an English script confusion reduction of -0.35 pp (95% CI: $[-0.65, -0.04]$). All intervals strictly exclude zero ($p < 0.05$).
- S_4 **vs.** S_1**:** demonstrates that sequence-level policy optimization provides statistically significant fine-grained gains in boundary fidelity (ΔSPER $= -0.07$ pp, 95% CI: $[-0.12, -0.03]$, $p < 0.05$) and English script confusion ($\Delta SCR_{en} = -0.09$ pp, 95% CI: $[-0.13, -0.06]$, $p < 0.05$). Paired bootstrap estimation for MER yields a mean difference of -0.03 pp (95% CI: $[-0.05, -0.01]$, $p < 0.05$).

To guard against potential intra-episode correlation across conversational sessions, we additionally conducted an episode-clustered block bootstrap ($B = 1{,}000$ resamples clustered across the 25 test podcast episodes). Clustered analysis confirms that the fine-grained advantage of S_4 over S_1 remains statistically significant: ΔSPER $= -0.07$ pp (95% CI: $[-0.15, -0.01]$, $p = 0.022$), $\Delta SCR_{en} = -0.09$ pp (95% CI: $[-0.11, -0.06]$, $p < 0.001$), and ΔMER $= -0.03$ pp (95% CI: $[-0.05, -0.01]$, $p = 0.008$). Because differing switch-point utterances are uniformly distributed across episodes (173 diffs across 20 distinct episodes), cluster aggregation stabilizes the ratio estimator across whole conversational sessions, yielding a tight interval consistent with utterance-level sampling.

5.3 Ablation Study: Gating vs. Policy RL

We conducted an extensive ablation on the development split (2,830 utterances) comparing: S_0 (base model), S_1 (synthetic SFT), S_2 (SFT with auxiliary script gate and inference logit biasing), S_3 (GRPO with MER-only reward, $\lambda_{scr} = 0.0$), S_4 (script-aware GRPO, $\lambda_{scr} = 0.5$), and S_5 (joint GRPO with script gate).

As shown in Table 2, adding an auxiliary token-level script gate (S_2) degrades performance relative to standard SFT (S_1), increasing MER from 18.92% to 19.14% and SCR_{en} from 5.35% to 5.75%. Error inspection reveals that greedy logit biasing forces premature script assignment at loanword prefixes, penalizing valid Devanagari assimilations.

Table 3: Monolingual retention evaluation on external benchmarks. All metrics in percentage (%).

Corpus	System	MER	WER	CER	SCR
Nepali (BHASAANUVAD) (1,000 utt, 1.95 h)	S_0 Base	20.80	30.73	22.52	8.71
	S_1 SFT	**18.09**	**29.06**	**19.22**	**6.95**
	S_4 GRPO	18.42	29.52	19.56	**6.95**
English (LibriSpeech) (1,000 utt, 1.50 h)	S_0 Base	**4.21**	**3.33**	1.59	**0.59**
	S_1 SFT	5.38	3.42	1.49	1.07
	S_4 GRPO	5.27	3.38	**1.47**	1.03

In contrast, sequence-level GRPO directly optimizes candidate probability mass without constraining autoregressive exploration. Comparing S_3 ($\lambda_{scr} = 0.0$) against S_4 ($\lambda_{scr} = 0.5$), both attain equivalent performance (18.89% MER, 25.53% SPER, 5.24% SCR_{en}) and produce identical hypothesis transcripts on dev. Inspection of training logs shows that because Stage 1 SFT already achieves 0.0200 MER on synthetic prompts, rollout candidate groups exhibit near-zero script variance during GRPO (over 75% of training mini-batches had zero within-group script reward variance). Consequently, the script term is inactive on the synthetic distribution, causing both runs to select the identical optimal checkpoint (step 50). While Ye and Vickers [7] found script rewards critical for penalizing unconstrained script hallucinations in generative audio LLMs, our ablation indicates that in an SFT-converged encoder–decoder model, sequence-level exploration is dominated by the primary transcription error rate, with the script reward contributing no measurable additional gain.

5.4 Monolingual Retention

A common drawback of code-switched adaptation is catastrophic forgetting of monolingual capabilities. We benchmarked S_0, S_1, and S_4 on two standardized external corpora:

Nepali (BHASAANUVAD, 1,000 utterances, 1.95 h). Rather than degrading, synthetic SFT (S_1) drops MER from 20.80% to 18.09% ($\Delta = -2.71$ pp), while S_4 stabilizes at 18.42% ($\Delta = -2.38$ pp vs. S_0). SFT provides the entire retention improvement by suppressing spurious Latin alphabet insertions in native Devanagari text (reducing SCR from 8.71% to 6.95%). Subsequent GRPO yields a minor trade-off of 0.33 pp in Nepali MER relative to S_1, as policy exploration slightly shifts Devanagari character boundaries to regularize English switch-points.

English (LibriSpeech test-clean [15], 1,000 utterances, 1.50 h). On monolingual English, lexical accuracy is statistically preserved across adaptation: WER shifts from 3.33% (S_0) to 3.42% (S_1) and recovers to 3.38% under S_4 (95% CI vs. S_0: $[-0.17, +0.30]$, $p > 0.05$). MER increases from 4.21% to 5.27% and SCR from 0.59% to 1.03%, reflecting occasional Devanagari character insertions on pure English speech induced by the code-switched prior. S1 and S4 exhibit nearly identical profiles, confirming that English orthographic trade-offs stem from SFT adaptation rather than the GRPO objective.

5.5 Qualitative Analysis and Error Inspection

Detailed inspection of transcriptions on the frozen test split illustrates how script-aware GRPO resolves boundary confusion:

- **Suppression of spurious transliteration:** In baseline model S_0, common English words such as "someone" and "brands" are frequently transcribed into phonetic Devanagari characters or split into mixed-script fragments (e.g., generating Devanagari roots with appended Latin plural suffixes). Under S_4, the sequence-level reward penalizes cross-script character boundaries, restoring continuous English orthography.
- **Stability at language transitions:** Transition regions between native Nepali matrix verbs and embedded English

noun phrases regularly induce insertion hallucinations in unadapted models. Synthetic SFT provides strong acoustic anchoring for code-switched prosody, while GRPO optimizes group-level sequence preferences, reducing boundary switch-point errors.

6 Limitations

While synthetic SFT combined with GRPO produces statistically reliable gains, four practical limitations remain:

1. **Modest RL gains relative to prior LLM studies:** While Kumar et al. [6] observed dramatic relative error reductions ($> 40\%$) from GRPO on synthetic data with decoder-only speech LLMs, our post-SFT GRPO gains on real spontaneous speech are more modest (-0.07 pp SPER, -0.09 pp SCR_{en}, -0.03 pp MER). We attribute this to: (i) structural architecture, where our FastConformer encoder–decoder model is already constrained by joint acoustic representations and SFT convergence, leaving less headroom than unanchored LLM decoders that suffer from massive insertion hallucinations; (ii) data overlap, as optimizing GRPO on the same 517 synthetic utterances previously fitted during SFT yields low reward variance across rollout candidate groups, rendering the script reward term inactive; and (iii) compute controls, where without multi-seed runs and an extended SFT compute control, the marginal 0.07 pp gain cannot be definitively isolated from additional training steps.
2. **Small synthetic training volume:** Training relies on 517 utterances (0.95 h), and Nepali speech is synthesized across two male timbres.
3. **English orthographic side-effects:** MER and SCR slightly increase on monolingual English despite unchanged WER.
4. **Demographic scope and external baselines:** The podcast benchmark predominantly reflects urban conversational discourse. While our study isolates adaptation stages across S_0–S_5, evaluating additional open-source models (e.g., Whisper-large-v3) on this new split will provide further external reference points.

7 Conclusion

We have presented an empirical benchmark and analysis of synthetic-to-real adaptation for Nepali–English code-switched ASR. Supervised fine-tuning on purely synthetic speech provides the primary transfer to spontaneous human conversational speech (-1.88 pp MER, -7.12 pp SPER). Sequence-level reinforcement learning using GRPO provides complementary fine-grained boundary regularization, outperforming token-level inference gating on dev and yielding small but statistically significant reductions in switch-point error rate (-0.07 pp) and English script confusion (-0.09 pp, $p < 0.05$) without degrading English lexical recognition accuracy (WER preserved at 3.38% vs. 3.33%), while improving native Nepali MER by 2.38 pp. Crucially, ablation reveals that script-aware rewards provide no measurable advantage over MER-only optimization in this SFT-converged encoder–decoder setting. The complete conversational benchmark (13,044 utterances, 15.45 h) is openly released on Hugging Face.

Code and configurations are open-sourced at `https://github.com/Firoj-Wise/nepali-english-codeswitch-asr`. The complete conversational benchmark (13,044 utterances, 15.45 h) is available on the Hugging Face Hub at `https://huggingface.co/datasets/SamaFiroz/nepali-english-codeswitch-podcasts`.

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